

Designing Embodied Agents for Impactful Human-Centred Risk Communication

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Abstract

Embodied agents are increasingly accessible to users and deployed in critical domains, which creates an urgent need for intuitive, human-centred risk communication to ensure users understand potential risks and infer appropriate action. My research explores embodied agents as novel interfaces for risk communication, investigating how these agents can intuitively convey risk information and facilitate risk alignment as an integral part of risk communication. I examine how using embodied agents to communicate risk impacts users' risk perception, decision-making, and understandability. Building on this work, my proposed research focuses on how agents can further support risk alignment by allowing users to specify and adjust their risk preferences through risk alignment interfaces. My dissertation contributes empirical insights, implications, and design considerations for human-centred risk communication in human-agent interaction and contributes to HCI by offering important insights for designing agents that prioritise risk comprehension and appropriate risk responses in complex environments.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

embodied agent, risk communication, affective visualisation design, risk alignment, human-agent interaction

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1 Motivation and Problem Statement

AI agents are increasingly integrated into daily life, ranging from LLM driven conversational agents to autonomous driving. They take on roles to support users in everyday tasks as assistants [17], facilitate or take over decision-making [7] and even act on the user's behalf [16, 19]. While doing so, they collect a vast amount of data

to inform such decision-making, to effectively collaborate, and to better align with the user [19]. When working with these systems, users are surrounded by risks—from external risks such as unexpected environmental changes to internal risks such as uncertainty in model predictions or failure in user alignment. In addition, users often lack an adequate mental model of the deployed agent, which can lead to misaligned expectations [22] or overreliance [6]. Whenever an AI agent is deployed, it is both an opportunity and crucial to inform users about risks (external and internal to the agent) and to support users' understanding of the risks at hand to ensure a great experience and safety. And yet, visualising and communicating risk in an intuitive, human-centered way in human-agent interaction has rarely been addressed so far.

The process of informing users about risks, to facilitate risk comprehension and to prompt an appropriate risk response comes with significant challenges. Risks are complex and often difficult to understand. They can be broadly defined as the possibility of a negative outcome as a result of being exposed to a specific hazard [47]. Thus, the risk of a specific hazard (whether it is external or internal) can be decomposed into two essential determinants, the *probability* of this hazard and its *consequence* [11, 36]. To support people's risk perception, traditional risk communication explores e.g. the use of anthropomorphic icons to better communicate risk statistics [37, 48] or the use of distressing and affect-inducing images to convey the consequences of risks [5] to elicit affective responses as a means to increase risk awareness. In line with affect-inducing risk communication, a recent paper on affective visualisation design highlights the potential of information visualisations to make viewers more engaged with the conveyed data, to educate, and to provoke [24] by appealing to the viewer's emotion. Here, affective visualisation design serves as a tool to increase expressiveness [3, 21, 23], and to boost user engagement [23].

Notably, contrary to AI-support systems or traditional information visualisations, *embodied* agents have a body and often share our environment. They can be perceived as social agents as a consequence of their embodiment [35, 41], appearance [31, 34] and continuously advancing interactivity, autonomy, and adaptability [20]. Based on these factors, my work demonstrates that *embodied* agents, in particular, introduce new possibilities for risk communication, i.e. to inform individuals *and* prompt emotional, cognitive, and reflective risk responses [38–40]. The affordances of embodied agents to convey risk information range from displaying traditional visualisations on graphical interfaces to offering a more immersive and engaging experience through the use of e.g. gestures, motion, or appearance-based changes. Visualising risk by leveraging an agent's physicality, anthropomorphism and perceived agency opens up new ways to make risk information more accessible, to

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amplify risk awareness, and to support understandability, since past research on AI-assisted decision-making has shown that providing users with explicit explanations is a double-edged sword in terms of reliance [2, 6, 8]. Moreover, research fields such as HRI and InfoVis rarely share best-practices to inform each others research, which has recently been identified as a major gap in HCI [43].

My research is motivated by a) the potential of embodied agents to communicate internal and external risks effectively through their physicality, anthropomorphism, and perceived agency and b) their versatile and increasingly expected ubiquitous role in everyday tasks, as social companions, health care assistants or collaborators. It further addresses the need to support an adequate mental model of the deployed agent and interaction context by communicating risks internal to the agent, and to ensure safety through enhanced risk perception, comprehension and risk alignment when interacting with such agents.

1.1 Research Objectives

My dissertation investigates important and novel approaches to human-centred risk communication through embodied agents. I identified four key research objectives (O1-O4). Specifically, my work explores how embodied agents can serve as both effective risk communicators (O1) and facilitators for risk alignment (O3). I develop interfaces and tools that support users in understanding and managing risks when interacting with such agents. My work provides insights into how embodied agents can enhance risk perception, understandability, and assisted decision-making (O2) to provide guidelines for human-centred risk communication in human-agent interaction (O4).

In the first phase of my research, I focused on exploring embodied agents as novel interfaces for risk communication, presenting embodied agents as *risk communicators* for internal and external risks. This phase resulted in two first-author publications, full paper and Late-Breaking Work, at CHI'24, with a third paper currently under review (CHI'25). I demonstrate how embodied agents can communicate risks through various affordances (O1), including conveying probabilities and risk consequences, and how it affects risk perception, perceived transparency and users' decision-making (O2). Which offers me first empirical insights to inform design considerations and implications (O4).

Building on this work, I propose the second phase of my research to focus on exploring embodied agents as *facilitators* for risk alignment (O3), an integral criteria of effective risk communication. Since risk communication is inherently bi-directional, this phase aims to investigate the design of intuitive interfaces that enable users to specify and adjust risk constraints for embodied agents, with the objective to allow users to clearly communicate their risk preferences (e.g. agent autonomy/ human control) in dynamic, risk-sensitive environments. The latter has the potential to be extended and applied to non-embodied agents.

In summary, the key research objectives of my PhD are as follows:

- O1:** Investigating agent affordances as novel interfaces to effectively communicate risk in a human-centered way, including
 - O1.1.** the communication of the probability of a risk and **O1.2.** the communication of risk consequences.

- O2:** Evaluating how an agent conveying risks affects the user's risk perception, decision-making and affective responses.
- O3:** Investigating how agents can facilitate risk communication and support users in defining and adjusting their risk preferences for effective human-agent alignment, i.e. risk alignment.
- O4:** Providing implications and guidelines to inform the design of human-centered risk communication through embodied agents.

Therefore, my dissertation contributes empirical insights, implications, and design considerations on human-centred risk communication within the embodied agent space. Collectively, these contributions move beyond traditional HAI and HRI challenges and push the boundaries of HCI by addressing real-world problems, filling critical gaps, and redefining the role of embodied agents in risk communication — as both communicators and facilitators in increasingly complex environments.

2 Current Status

2.1 Uncertainty Visualisation and Embodiment in Robot-Assisted Decision-Making

This study contributes to O1.1., i.e. how to visualise *the probability of a hazard* when interacting with an embodied agent [40]. We contextualise this study in a healthcare scenario due to the expected implementation of embodied agents in health-related contexts [9, 13, 14], its high-risk nature and the critical need for effective risk communication [26]. In this work, the agent visualises its confidence level in the quality of a pill box (through probability values) during a collaborative assessment and decision-making task, and thus conveys the associated risk.

In detail, I a) investigate different visualisation modalities for the embodied agent to convey probabilities by integrating best-practices from HRI and InfoVis [43] (O1.1.) and b) its effects on users' decision-making, perceived transparency, and trust in the agent (O2) in a high-stakes scenario. Notably, "the relationship between uncertainty estimates and trust in automation is a relatively unexplored idea" [4]. Prior research further suggests that the communication of uncertainty can enhance transparency, calibrate trust and potentially support assisted decision-making [4, 30]. Yet, visualising probabilities through an embodied agent and its effects on assisted decision-making, perceived transparency and trust in a high-stakes context has rarely been investigated so far.

Method. We conducted a 3×2 mixed design lab study. Our work considers two affordances for the probability visualisation, a graphical user interface (GUI) and an EMBODIED visualisation. For the first approach, we applied a part-to-whole icon array as an established visualisation technique for probabilities [1, 26, 28], confirmed through our manipulation check. The visualisation was displayed on a simple screen attached to the agent through a 3D printed case. The screen showed a GUI via a Flask webapp, which communicated with the agent to convey the respective probability visualisation. For the second approach, we used the agent's physicality to map the probability values onto the agent's movements, which we termed an EMBODIED visualisation. To inform and guide our EMBODIED design and the implemented trajectory function, we followed research on hesitation gestures in HRI [18, 27] and drew on insights

related to visualising motion intent [25, 29] and expressive motion behavior [10]. We further manipulated the level of confidence, i.e. probability value, to make inferences about the user’s understanding and perception of the magnitude of risk conveyed by the agent.

Results and Contribution. Our findings reveal that the type of uncertainty visualisation significantly affects users’ decision-making. We show that the agent’s conveyed confidence level significantly impacts the user’s perceived transparency of the embodied agent. In addition, we discuss a potential trade-off between an intuitive or informed decision-making when visualising uncertainty through an embodied agent and how conveying a fully confident agent can be a double-edged sword in an assisted decision-making task. Besides empirical findings, our work contributes critical considerations for risk visualisations in high-stakes, agent-assisted decision-making tasks and new avenues for future research (O4).

2.2 Exploring an Agent’s Face as Affective Visualisation in Risk Communication

This work serves as an initial step to explore O1.2., i.e. how to visualise *risk consequences* through an embodied agent [39]. A common inhibitor for risk communication is that it is extremely difficult to envision the *consequences* of (often) abstract risks, and *how they would feel* once experienced [45]. Risk communication research shows that emotions play a critical role in how we perceive and understand risks [12, 33, 46], acknowledging “risk as feelings” [42]. These “risk feelings” become particularly relevant when communicating risk consequences. The perceived consequence of a risk is determined by two factors: the perceived severity of a consequence and affective responses triggered by the hazard [47].

I therefore propose that the communication of risk consequences through embodied agents has to support people’s *understanding of the consequence*, but also *make them feel*. Inspired by the challenge to “transform complex, cold data into vivid, affective representations” and the need to explore new technologies for affective visualisation design [24], my work explores the use of an embodied, humanlike agent as a visualisation interface to communicate risk. In specific, my work examines the potential of an agent’s face to communicate emotion as a foundation of effective risk communication *and* to inform about the negative consequences of risks, including their severity [39]; a concept I coined FACEVIS [39].

Method. As a first step towards these objectives, we conducted a design workshop with 9 experts from HCI related fields to identify challenges, opportunities, and design strategies for using an agent’s face as an interface for risk communication. A design workshop is a suitable practice for exploratory work in HCI, to facilitate idea generation, and to generate visualisation designs [15]. The design workshop included both facilitated brainstorming activities and group discussions to unveil potential challenges and opportunities of FACEVIS. The workshop focused on visualising risks based on data related to health & well-being, environmental issues and social issues [24]. For instance, we asked experts to visualise different types and levels of air pollution using an agent’s face as their visualisation canvas.

Results and Contribution. From the workshop, we collected a total of 23 design outcomes. Based on a group discussion, we identified challenges and opportunities for using a humanlike embodied agent

for risk communication. We further implemented several designs as a first prototype¹, using the Furhat robot [32], a popular humanlike conversational agent. The design workshop forms a first step to gain a preliminary understanding of how an agent’s humanlike face can be used to map and convey risk consequences and to push the boundaries of information visualisation to communicate real-world risks “face-to-face”, which I evaluate in the following study.

2.3 Affect-inducing Risk Communication through an Embodied Humanlike Agent

Informed by the design workshop, we implemented refined designs on a real-world social agent, i.e. Furhat [32], to evaluate the effectiveness of using an embodied humanlike agent as a visualisation interface to communicate risk consequences guided by O2. Driven by the increasing implementations of social agents in health-related domains, we focus this work on health risk scenarios, e.g. air pollution and its consequences to human health. In detail, this study explores (i) the effectiveness of using an embodied humanlike agent to communicate the consequences associated with risks external to an agent (but relevant to the user); (ii) how it impacts affective responses from viewers, empathy, self-reflection and motivation to act on the conveyed risk; (iii) potential application contexts. The study is submitted and under review for CHI’25.

Method. We conducted a mixed-method user study involving 28 participants to assess changes in participants’ risk perception, self-reflection, empathy, and motivation to act pre and post exposure to the agent visualisation. In detail, the study focuses on two hazard scenarios: LDL cholesterol and PM2.5 (as an indicator for poor air quality). Both scenarios were selected based on their real-world significance as health risks and proximity to the workshop stimuli. I developed a browser-based graphical user interface through which participants were able to manipulate data values for each hazard scenario using a simple slider. As participants adjust the slider and thus data values, the corresponding visualisation is updated in real-time and conveyed through the agent, visualising the respective consequences through the agent’s face.

Results and Contribution. Our findings reveal that using an embodied humanlike agent as an interface to visualise risk consequences facilitates data comprehension, heightens risk perception, elicits empathy, and motivates behavioural change by making risks relatable and emotionally resonant. We discuss the potential of using such interfaces for risk communication in public spaces, health campaigns, and education — domains in which social agents are expected to be deployed to inform and support humans. Beyond, this approach could inspire research involving other agents, such as virtual avatars, to further explore the impact of human-centered risk communication through appearance-based changes in digital environments. We provide design considerations, key takeaways and future directions for advancing this important and novel pathway in human-centred risk communication.

¹<https://sites.google.com/view/facevis/home>

3 Proposed Research Direction

3.1 AI Agents as Facilitators for Risk Alignment

My previous research projects explore embodied agents as risk communicators, visualising both the probability of risks and their consequences for users. However, risk communication is a bi-directional process. In my next research, I plan to build on this work by examining how AI agents can act as facilitators for risk alignment guided by O3, providing interfaces that enable users to input and adjust their risk preferences based on the probability and consequences of potential risks. This shift moves the focus from the agent's ability to communicate risk to risk alignment between the agent and the user, proposing risk alignment as a key criteria for effective risk communication in human-agent interaction. Human-Agent Alignment is an emerging topic in HCI, since AI agents are gaining more autonomy in their roles based on their continuously advancing capabilities [16, 17]. As AI agents take on critical responsibilities, it becomes vital for users to be able to specify clear constraints, preferences and boundaries. Specification alignment describes the need to design for alignment opportunities in a way that users can effectively communicate not just outcome objectives but constraints to AI agents [44]. This becomes particularly essential in scenarios with varying risks involved.

To achieve effective risk alignment, it is essential to design tools and interfaces that allow users risk specification alignment, i.e. to specify and adjust their preferences based on the probability and consequence of risk within a scenario. Such interfaces would enable users to align risk constraints in ways that ensure the AI agent operates transparently and according to their oversight needs. The challenge lies in understanding how to operationalise risk alignment, and which risk factors most influence user decisions when collaborating with AI agents in different scenarios - which eventually informs the design for respective interfaces.

In the second phase of my PhD, I aim to investigate tools and interfaces for risk alignment, focusing on how to effectively support users in defining risk boundaries (probability and consequence) when interacting with AI agents. I identified three open challenges:

- C1:** How can we design intuitive interfaces that enable users to specify and adjust risk constraints for AI agents, allowing them to clearly communicate their preferences in dynamic, risk-sensitive environments?
- C2:** How does the presentation of risk (through visualisation or interaction tools) affect users' ability to effectively align AI agent behaviour with their own risk preferences?
- C3:** What factors (probability, consequence, risk context) do users prioritise when defining boundaries for AI agent autonomy, and how do these factors influence their decision-making process?

4 Conclusion

My dissertation focuses on how embodied agents communicate internal and external risks and how this affects users' perception, decision-making, and understanding. Informed by multidisciplinary insights, my research aims to make risk information intuitive and actionable through embodied agents. It contributes empirical findings and design considerations for human-centred risk communication in human-agent interaction, and aims to offer guidance for designing

agents that support risk comprehension and appropriate responses in complex environments.

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